

Continuous Water-Quality Monitoring with a Low-Cost Autonomous Sensor Buoy

Protocol ID	schmidt_2018
Sector	Molluscs & echinoderms (blue mussel)
Usability tier	Tier 4 — Practitioner (designed for non-scientific staff to operate)
What it monitors	Dissolved oxygen at the farm site, logged continuously — a water-quality benefit indicator. Temperature (at several depths), salinity, and water depth are recorded as supporting context.
Source paper	Schmidt, W., Raymond, D., Parish, D., Ashton, I.G.C., Miller, P.I., Campos, C.J.A. & Shutler, J.D. (2018). Design and operation of a low-cost and compact autonomous buoy system for use in coastal aquaculture and water quality monitoring. <i>Aquacultural Engineering</i> , 80, 28–36.
Associated tools	Part of a bundle: this normalized protocol, a field datasheet (a deployment / calibration / maintenance log), and a stats workbook. (No separate calculator applies to this protocol.) Download links are added once the bundle is hosted.

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Adapted into a general, gear-agnostic method for AquaPIP. Open-access CC BY (Elsevier).

Introduction

Near-shore waters change fast — over a day, a week, a season — and monthly grab samples miss most of it. This protocol brings continuous, automatic water-quality monitoring within reach of a small shellfish operation using a low-cost sensor string on a buoy that non-scientific staff can run.

This paper reports the design and two deployments (14 months in total, a coastal bay and an estuary) of a sub-£5,000 autonomous buoy carrying temperature, salinity, dissolved-oxygen and pressure sensors that log automatically and are downloaded on routine visits. The authors highlight that the system was built to be operated and maintained by non-scientific staff while still producing data good enough to confirm site suitability for mussel and lobster culture — and to flag conditions that favour a toxin-producing alga. The aim is a repeatable, low-effort record of water quality at the farm.

Parameters measured

This protocol records two kinds of measurement. Monitoring linked to a selected indicator (green) is the data that demonstrates a MEL benefit this protocol targets — here, dissolved oxygen at the farm site. Context monitoring (grey) captures the conditions around that data — temperature, salinity, the instrument's depth — so the dissolved-oxygen readings can be interpreted and stood behind; it is recorded but is not itself tied to a selected indicator. Each parameter below is listed with its unit and a working definition.

■ monitoring linked to a selected indicator ■ context monitoring

Dissolved oxygen — the water-quality indicator

- **Dissolved oxygen (mg L⁻¹)** — DO concentration at the farm site, logged continuously by the sensor at a fixed interval; the water-quality benefit indicator this protocol targets. Read it for daily and seasonal patterns (e.g. bloom supersaturation; night-time draw-down).

Physical context — context monitoring

- **Water temperature (°C)** — logged at several depths (e.g. ~0.6–1.1 m) to detect stratification, which helps interpret dissolved oxygen and biological activity.

- **Salinity (PSU)** — derived from a conductivity sensor (see Methods referenced); tracks river and tidal influence.
- **Water level / depth (m, from pressure)** — confirms the actual depth and vertical orientation of the instrument string, and flags when the string has floated to the surface (those readings are then excluded).
- **Sensor-vs-reference accuracy (RMSE, bias)** — QA: agreement of each sensor against a calibrated reference (e.g. a hand-held CTD) taken on maintenance visits.
- **Biofouling & battery state** — operational notes recorded at each visit (fouling growth, 'bad battery' events).

Requirements

- **Personnel [estimate]:** non-scientific staff can deploy and maintain the buoy; one person with a boat and a low-power winch (like those used to lift mussel strings) for routine visits, with occasional partner support for calibration and reference readings.
- **Time per visit [estimate]:** around [20 min] to anchor onto the main mooring, lift the instrument string, download the data to a handheld shuttle, clean biofouling, and return the string to the water — plus a reference (CTD) reading where available.

Sampling frequency & timing

- **Framework guidance.** The MEL framework sets the mollusc dissolved-oxygen indicator (1.2.1) at weekly / monthly / seasonal, with diurnal measurements over several days — i.e. frequent enough to capture daily cycles.
- **What the paper reports.** This paper logs every 10 minutes continuously for months at a time (two deployments totalling 14 months), downloading data on monthly visits — comfortably capturing diurnal, weekly and seasonal variation.
- **Recommended to detect an effect.** Log at a short fixed interval (e.g. every 10 minutes) across at least a full seasonal cycle, with routine (e.g. monthly) maintenance visits, so the diurnal day/night DO swing a monthly grab would miss is captured.

► One buoy describes the site; a farm effect needs a comparison

The framework samples dissolved oxygen “within the farm area and adjacent non-farm areas.” A single buoy characterises conditions at the farm site well; to attribute a difference to the farm rather than the wider waterbody, add a comparison reading at a nearby non-farm location [bracketed], or a reference instrument, on the same schedule.

Materials

Materials are roles to fill, not products to buy. The source used specific loggers, mooring parts, and software; substitute equivalents that fill the same roles.

► Gear varies site to site — use your best judgement

The source used specific equipment (a set of HOBO U-series loggers, a 100 L float with a concrete-filled-tyre anchor, and particular software). Availability, budgets, and site exposure differ from farm to farm. Substitute an available sensor string, mooring, shuttle, and software that fill the same roles — nothing below is mandatory as a specific product.

- **Shared for the deployment:** a moored buoy/float with an anchor sized to the site's exposure; an instrument string carrying temperature (at several depths), conductivity/salinity, dissolved-oxygen and pressure sensors, each with an internal logger and battery; antifouling guards for the dissolved-oxygen and

conductivity sensors; a handheld data shuttle; a calibrated reference instrument [e.g. a hand-held CTD]; a boat with a low-power winch; GPS; and ID tags with contact details on the string.

Methods

Single-method protocol — a moored autonomous sensor string that logs continuously, serviced on a routine cadence, with a quality-control pass before the data are used.

Deploy and operate the autonomous sensor buoy

↳ *Linked indicator: dissolved oxygen concentration at the farm site — match: **Exact**. Full mapping in Appendix A.*

Materials. a moored buoy/float and site-appropriate anchor; a vertical instrument string (temperature at several depths, conductivity/salinity, dissolved oxygen, pressure) with internal loggers; antifouling guards; a handheld data shuttle; a calibrated reference [e.g. CTD]; a boat with a low-power winch; GPS.

- **Preparation.** Choose the parameters to log [here temperature at several depths, conductivity/salinity, dissolved oxygen, and pressure]. Calibrate the dissolved-oxygen sensor before deployment following its 3-step procedure [100 % in water-saturated air; a temperature-equilibration step; 0 % in a sodium-sulphite solution]. Set every sensor to log on a common short interval [e.g. every 10 min] with clocks synchronised, and fit antifouling guards to the dissolved-oxygen and conductivity sensors.
- **Deployment.** Moor the buoy following a reliable local mooring design; hang the instrument string vertically with a bottom weight to keep it oriented, and include a pressure sensor to record depth and flag surfacing. Build the mooring in two parts so the light instrument chain can be lifted without moving the main mooring. Record position and each sensor's depth.
- **Maintenance.** Visit periodically [e.g. monthly]: lift the instrument string, download all data to the handheld shuttle, scrub biofouling from the sensors, and — where possible — take a reading with the calibrated reference to check accuracy. Note battery and fouling state at each visit.
- **Data processing & QC.** Convert conductivity to salinity in software (see Methods referenced). Apply simple quality-control rules: drop readings outside sensible ranges; discard the first couple of hours after each deployment or cleaning; remove single-point spikes; cut the gradual biofouling drift before each cleaning; and exclude readings when the pressure sensor shows the string has surfaced.
- **Analysis.** Report dissolved oxygen (and the context parameters) as time series. Where a reference reading was taken, quantify sensor accuracy (RMSE and bias). Read dissolved oxygen for daily and seasonal patterns and against thresholds relevant to the stock.

Expect these failure modes and constraints (and manage them)

Each constraint is grouped under the method or the scope it affects, with how to manage it.

Sensor buoy — deployment & data

- **Biofouling** — the single biggest limiter of accuracy and deployment length; use antifouling guards, clean the sensors at every visit, and keep the QC rules that cut fouling drift before each cleaning.
- **Storm damage / mooring failure** — instrument strings can tangle around the float, and a surface buoy can part from the mooring in a storm; use a robust mooring and the pressure-sensor surface-flag, and move any spliced rope away from split pins/shackles.
- **Sensor drift** — dissolved-oxygen and conductivity sensors drift over time; recalibrate periodically (the dissolved-oxygen sensor in particular).
- **Reference-instrument dependence** — accuracy checks are only as good as the reference; confirm the reference instrument is calibrated and working before trusting a comparison (the source's first reference instrument was faulty and its readings unusable).
- **String orientation** — the string can rotate and tangle around the float and lose vertical orientation; the pressure sensor records actual depth so affected readings can be identified and excluded.

Scope & interpretation

- **Physico-chemical only** — the buoy characterises dissolved oxygen, temperature and salinity; it does not directly measure microorganisms or harmful algae. Algal risk is inferred from physico-chemical proxies (e.g. stratification), so direct plankton detection stays a separate, external step — which is why the microorganism indicator is not mapped here.
- **Site vs. farm effect** — one buoy describes conditions at the site; attributing a change to the farm rather than the wider waterbody needs a non-farm comparison reading.

Data handling

Use a consistent labelling convention: site — sensor — deployment — date. The readings live in the sensor CSV downloads; the field log holds deployment metadata (positions, sensor depths, service visits, calibration, and fouling/battery state), and the workbook's data-entry columns mirror that log exactly. Archive raw sensor downloads and reference (CTD) casts against their deployment IDs. [Insert your own mooring diagram / site photo] for the service crew.

Methods referenced from other sources

Two steps rely on established external tools rather than the source article. Resolve and verify a stable link for each before relying on it.

- **Conductivity → salinity conversion** — turns the conductivity readings into salinity (PSU) in the Data-processing step, using conductivity ratio, pressure and in-housing temperature. Soetaert, K., Petzoldt, T. & Meysman, F. (2015). *marelac: Tools for Aquatic Sciences (R package; 'convert_RtoS')*. <https://CRAN.R-project.org/package=marelac>
- **Real-time quality-control rules** — the range, stabilisation, spike and drift rules applied in the Data-processing step. U.S. Integrated Ocean Observing System (2015). *Manual for Real-Time Quality Control of In-situ Temperature and Salinity Data (QARTOD)*. <https://ioos.noaa.gov/project/qartod/>

Appendix A — MEL objective table

Reference: how the monitoring maps to the MEL framework. The method links to a MEL Objective, the metric it produces, and how closely that metric matches the objective (the per-method “Linked indicator” line above is the short form of this table). The premise: the “how” of monitoring can differ across studies as long as the metrics are the same. Framework codes appear only here.

Goal	Monitoring method	MEL Objective	Metric(s)	Match
GOAL 1 Mollusc and echinoderm farming contributes to a reduction of excess organic nutrients in the local environment (dissolved oxygen at the farm site is the water-quality indicator).	Autonomous sensor buoy — continuous dissolved-oxygen logging	Objective 1.2 (Ind. 1.2.1) — Unfed farmed molluscs support a reduction in symptoms of eutrophication; dissolved oxygen concentration at the farm site	Dissolved oxygen concentration (mg/L) at the farm site, logged continuously (diurnal cycle captured)	 Exact

Match scale. How closely the method's metric matches the objective's.

 **Exact** — measures the objective's metric as specified.
 **Partial** — measures a proxy for, or subset of, the metric, or needs an added condition such as a control site.
 **Indirect** — gives relevant evidence toward the objective (a native species present/active) but not the specified metric.

Not mapped (context). Temperature (at several depths), salinity, and water level/depth are recorded to interpret the dissolved-oxygen readings (stratification, river/tidal influence, instrument orientation) but are not themselves mapped to a Water Quality indicator. Microorganisms and harmful algae are not directly measured — algal risk is inferred from physico-chemical proxies, so that indicator is out of scope for this protocol.

Appendix B — Provenance & practitioner decisions

- **Adapted from:** Schmidt, W., Raymond, D., Parish, D., Ashton, I.G.C., Miller, P.I., Campos, C.J.A. & Shutler, J.D. (2018). Design and operation of a low-cost and compact autonomous buoy system for use in coastal aquaculture and water quality monitoring. *Aquacultural Engineering*, 80, 28–36. <https://doi.org/10.1016/j.aquaeng.2017.12.002> (Open access, CC BY).
- **Indicator set:** The source row's mapped indicator — dissolved oxygen at the farm site — is monitored here. The microorganism / harmful-algae indicator the study touches on was not mapped: the buoy infers algal risk from physico-chemical proxies and does not directly detect plankton, so that indicator is out of scope for this protocol (monitored-only rule).
- **Decisions left to the practitioner [bracketed]:** these are choices this protocol deliberately leaves open for the site — not gaps the original authors reported. Set them locally: which parameters and how many depths to log; the logging interval [e.g. 10 min]; the mooring design and anchor sizing for the site's exposure; the maintenance interval; the reference instrument and how often it is used; the dissolved-oxygen thresholds relevant to the stock; and whether a non-farm comparison reading is taken. A full read of the source narrowed these but does not remove them; do not silently fill a value to look complete.